
Toward Mechanical Scientific Models: Fractal Custody and Governed Agent Experiments

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Abstract

The *Mechanical Scientific Method* treats scientific computation as a sequence of governed state transitions in which evidence, transforms, and bounded claims remain inspectable rather than collapsed into model prose. We propose the term *Mechanical Scientific Model* for a computational model class that satisfies explicit requirements: explicit evidence state; governed and versioned state transitions; separation of deterministic transforms from probabilistic model outputs; preservation of positive, null, negative, failed, blocked, timeout, abstention, and contradictory outcomes; explicit claim ceilings; forward and reverse evidence traceability; and successor-state recording rather than silent history rewriting. *Fractal Custody Objects* (FCOs) and a *Fractal Custody Graph* (FCG) provide the custody substrate for binding those transitions. HydraDG is the primary experimentally evaluated implementation examined here: it preregisters hypotheses, freezes case manifests and scorers, records raw prompt and response hashes, and preserves terminal states without silent substitution. We report two preregistered HydraDG studies (EXP-008, EXP-009) comparing flat prose versus structured FCG retrieval for failure-prevention endpoints; both terminate as *underpowered*, so confirmatory treatment-effect claims are not supported while null, negative, and failure evidence remains in custody. We do not claim that Mechanical Scientific Models are an established external field term; we offer them as a proposed model class and report bounded custody behavior from one implementation.

1 Introduction

Large language model agents are evaluated on task success, benchmark scores, and human preference [2, 3]. Less often preserved are the conditions under which experiments fail: swapped models, stale runtime selection, partial matrix execution, or JSON outputs that never parse. When negative and null cells disappear from the public record, later work cannot distinguish *no effect* from *no measurement*.

This paper separates five layers that are often conflated in agent-evaluation reporting:

- **Method** — the *Mechanical Scientific Method*: governed transitions among explicit evidence states rather than narrative collapse of runs into conclusions.
- **Model class** — *Mechanical Scientific Models* (proposed here; not an established external terminology): computational systems whose state transitions are mechanically exposable and failure-complete.
- **Custody substrate** — FCO/FCG receipts that bind source bytes, transforms, derived evidence, bounded claims, and successor state.
- **Primary implementation** — HydraDG, the experimentally evaluated implementation whose preregistered EXP-008 and EXP-009 results are reported here.
- **Systems validation and companion exemplars** — HydraLamp perturbation receipts and HydraDG scoring on local Apple Silicon (Mac Studio class, 32 GB RAM); companion

39 implementation lineages are excluded from primary inference for EXP-008/009 and are
40 not part of the reviewed source set for this submission.

41 HydraDG instantiates the proposed model class through a custody-first experimental contract. Every
42 actor—human operator, frontier agent, local model runtime, or deterministic script—must emit or
43 be represented by a handoff receipt before its output is promoted. Receipts bind actor identity,
44 host, repository state, input hashes, raw output hashes, evidence class, claim ceiling, and signature
45 state (hashes are identity commitments, not digital signatures unless an authorized signing operation
46 occurred). Probabilistic model outputs are never treated as authority; they are evidence objects
47 subject to deterministic scoring and graph append rules.

48 This paper reports terminal results from the IC Failure Learning structured-retrieval falsification lane
49 executed in HydraDG. Primary numbers are reverse-traced to frozen preregistered verdict JSON re-
50 cepts retained as internal anonymized custody artifacts. We do *not* claim that structured FCG con-
51 text improves failure prevention: EXP-008 and EXP-009 closed as *underpowered* ($n_{\text{paired}} = 2$,
52 $\text{discordant} = 0$) with bounded conclusion that the effect is not established. Exploratory secondary
53 patterns from companion implementations are not promoted to primary findings for those experi-
54 ments.

55 2 Related Work

56 **Agent evaluation.** Benchmark suites measure capability but rarely require publication of failed
57 runs, malformed generations, or infrastructure blocks [2, 3]. HydraDG complements benchmarks
58 with mandatory negative capture at the custody layer.

59 **Retrieval and structured context.** Retrieval-augmented generation and graph-augmented memory
60 systems assume structured context helps [1, 4]. Falsification requires frozen contracts, explicit null
61 retention, and power-aware verdicts; HydraDG enforces both.

62 **Local open-weights evaluation.** Frozen EXP-008/009 cells use qwen3:1.7b and
63 qwen2.5-coder:7b only. Historical LongMemEval track-03 retrieval ablations (local qwen2.5-7b
64 and deepseek-r1-14b) are excluded from primary inference and are not promoted here. No
65 frontier API model executed a preregistered primary scorer in this submission.

66 **Provenance, integrity, and reproducibility.** W3C PROV-O and workflow provenance systems
67 represent process and research-object lineage [12, 13], while append-only Merkle audit logs pro-
68 vide established tamper-evident integrity mechanisms [11]; HydraDG does not claim to invent these
69 primitives. Recent work surveys evidence tracing and execution provenance for LLM agents [15],
70 and separate work argues for preserving negative ML results rather than selective omission [14].
71 Preregistration, FAIR principles, and nanopublication-style lineage further support reproducibility
72 [5–7]. HydraDG integrates rather than replaces these foundations: probabilistic outputs, determin-
73 istic transforms, malformed cells, failures, abstentions, blocked states, and underpowered termina-
74 tions remain typed evidence with explicit claim-promotion ceilings enforced via FCO/FCG handoff
75 receipts. Companion custody-framework preprints (deferred to camera-ready de-anonymization) de-
76 fine FCO/FCG formally; they establish framework provenance, not the HydraDG empirical results
77 reported here.

78 3 Framework

79 3.1 Mechanical Scientific Models

80 We propose the term *Mechanical Scientific Model* for a computational scientific system whose inter-
81 nal evidence-state transitions are mechanically exposable rather than inferred post hoc from model
82 prose. We do not claim that this term is already established in the literature; it names a proposed
83 model class whose required properties can be checked against an implementation.

84 Let S_t denote the governed evidence-and-claim state at step t . A Mechanical Scientific Model
85 advances by governed transforms

$$S_t \xrightarrow[E_t]{T_t} S_{t+1},$$

86 where T_t is a versioned, auditable transform (deterministic scorer, ingest pass, governed model call,
87 or tool invocation) and E_t is the evidence bound to that transition. Every transition must emit or
88 bind:

- 89 1. source or upstream evidence;
- 90 2. transform, tool, or model identity (with deterministic vs. probabilistic output separation);
- 91 3. derived evidence or parser state;
- 92 4. a bounded claim with an explicit ceiling; and
- 93 5. an artifact or state delta recorded as successor state rather than silent history rewriting.

94 Required properties include: explicit evidence state; governed and versioned state transitions; deter-
95 ministic vs. probabilistic output separation; preservation of positive, null, negative, failed, blocked,
96 timeout, abstention, and contradictory outcomes; explicit claim ceilings; forward and reverse ev-
97 idence traceability; and successor-state recording. Figure 1 situates the Mechanical Scientific
98 Method, FCO/FCG custody, the proposed model class, HydraDG as the evaluated implementation,
99 companion exemplars excluded from primary inference, and execution components used in this sub-
100 mission.

101 3.2 Custody objects

102 An **FCO** (*Fractal Custody Object*) materializes one bounded transformation: e.g., a model call,
103 scorer pass, or ingest operation. An **FCG** (*Fractal Custody Graph*) append records directed edges
104 between FCOs with hash-linked roots [11]. FCO/FCG provide the custody substrate on which
105 Mechanical Scientific Models record transitions; HydraDG is one implementation that preserves
106 failures, nulls, and malformed outputs in custody (a failure-complete *behavior*, not a redefinition of
107 FCO/FCG). Claim ceilings (exploratory mechanistic falsification, descriptive only, etc.) are declared
108 at preregistration and enforced at promotion time.

109 3.3 Experimental freeze

110 Each experiment declares:

- 111 • frozen case manifest and prompt-contract hashes;
- 112 • conditions (e.g., flat prose vs. structured FCG);
- 113 • models with expected runtime digests;
- 114 • primary endpoint and aggregation rule;
- 115 • explicit gates for confirmatory vs. exploratory claims.

116 Execution emits START_RECEIPT, per-cell raw outputs with prompt/response SHA-256, parser state,
117 and terminal receipts. Integrity failure (duplicate keys, missing dependencies, silent model substi-
118 tution) blocks promotion; scientific failure (timeout, abstention, malformed JSON) is retained as
119 terminal evidence.

120 3.4 Governed local execution

121 Where preregistered, local model calls route through a governed bridge that records selection age,
122 swap posture, and resolved digest. Direct runtime invocation and governed invocation are separate
123 evidence classes; one is not silently labeled as the other. EXP-008/009 executed on an Apple Sili-
124 con Mac Studio class workstation (32 GB RAM) with HydraDG scoring and HydraLamp systems-
125 validation receipts on the same host class.

126 3.5 Evaluation model policy

127 All preregistered scientific endpoints in this submission use small local open-weights models only
128 (qwen3:1.7b, qwen2.5-coder:7b). Frontier coding agents assisted manuscript preparation only;
129 they did not execute frozen primary scorers or replace verdict-bound cells.

F1: Proposed hierarchy (conceptual; not treatment-effect evidence)

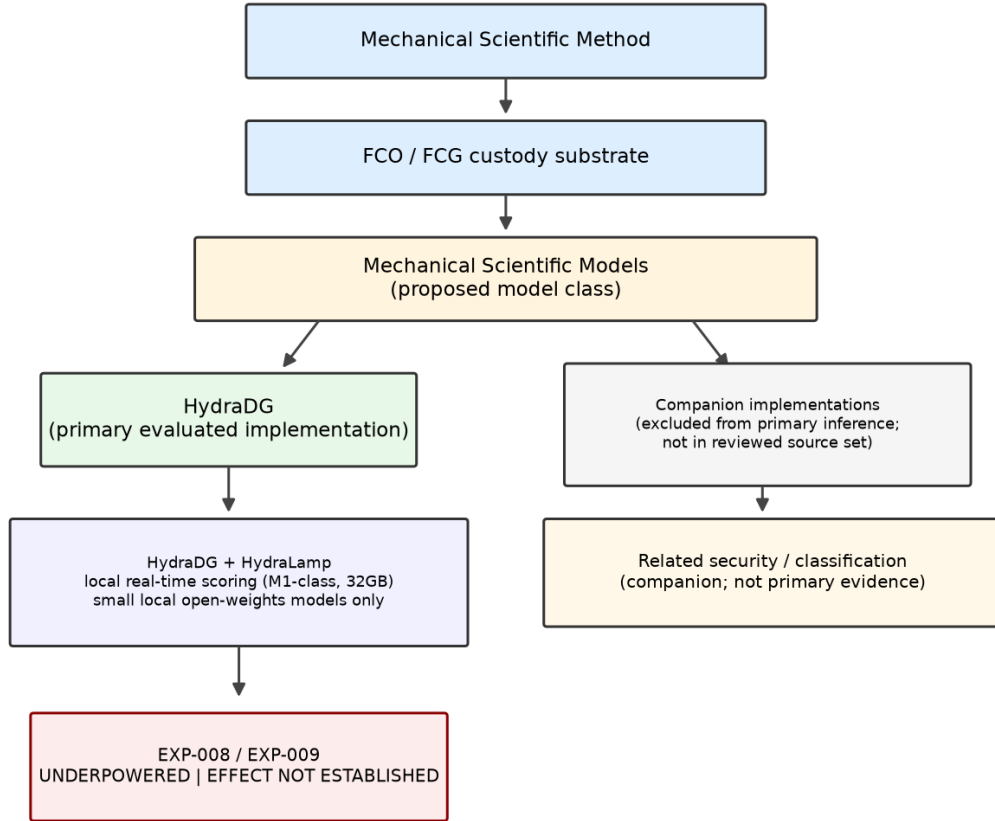


Figure 1: Proposed intellectual hierarchy (conceptual; not empirical effect evidence). HydraDG is the primary experimentally evaluated implementation; companion implementations are excluded from primary inference and are not available in the reviewed source set; EXP-008/009 remain *underpowered*.

130 4 Experimental Setup

131 4.1 Task family

132 The IC Failure Learning suite uses synthetic agent-failure vignettes with preregistered endpoints
 133 such as E06 (prevents exposure of out-of-vault media). Cases are drawn from a frozen CASES . jsonl
 134 manifest; each case receives multiple replicates per condition.

135 4.2 Conditions and models

136 **C0 (FLAT_PROSE)**: failure-handling guidance as unstructured text.

137 **C1 (STRUCTURED_FCG)**: the same semantic content composed through a frozen FCG retrieval
 138 contract.

139 EXP-008 and EXP-009 used local models qwen3:1.7b and qwen2.5-coder:7b under identical
 140 case and scorer contracts on private M1-class hardware (32 GB RAM). Thinking/reasoning modes

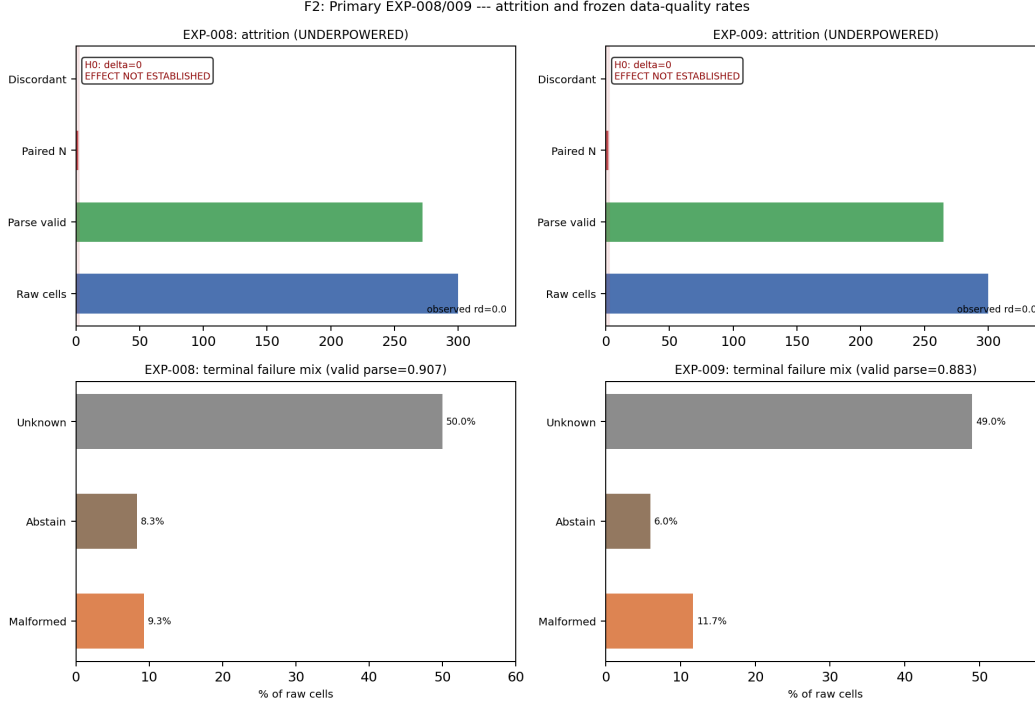


Figure 2: Primary EXP-008/009 outcomes from frozen verdict JSON: top panels show attrition (raw → paired); bottom panels show terminal failure rates (malformed, abstain, unknown). H0 reference shown; UNDERPOWERED — effect not established (not proof of null).

were held consistent with preregistration. Primary aggregation: majority of three replicates per case, then paired comparison at the model-stratum level. The preregistered confirmatory aggregation operates at the paired model-stratum level, yielding $n_{\text{paired}} = 2$; the 300 raw execution cells are pre-aggregation replicates and do not constitute 300 independent confirmatory treatment pairs. Historical LongMemEval track-03 retrieval ablations under the same local-only policy are excluded from primary inference and are not promoted here.

4.3 AI and agent methodology disclosure

Frontier coding agents assisted with repository tooling and manuscript preparation; they are not listed as authors. All scientific endpoints come from preregistered local model executions with hashed prompts and responses. Routine editing does not constitute a reported experimental intervention.

4.4 Power and claim gates

E06 endpoints are known limited-power under the frozen case set. Confirmatory ordering claims require preregistered discordant pairs; exploratory secondary statistics are quarantined from primary promotion.

5 Results

Table 1 summarizes terminal verdicts for preregistered studies. Figure 2 decomposes the same receipts into attrition counts and data_quality rates ($n_{\text{raw}} = 300$ per study). Both experiments executed the full raw matrix with complete custody append.

EXP-008. Structured FCG retrieval did not yield promotable confirmatory evidence on E06 under the frozen design ($n = 2$ paired cases per model stratum at aggregation scope). Non-zero malformed and abstention rates are preserved in custody.

Table 1: Terminal preregistered studies (HydraDG IC Failure Learning). *Underpowered* denotes insufficient paired evidence for confirmatory promotion, not proof of null effect.

Study	Primary verdict	Raw cells	Valid parse rate
EXP-008	UNDERPOWERED	300	0.907
EXP-009	UNDERPOWERED	300	0.883

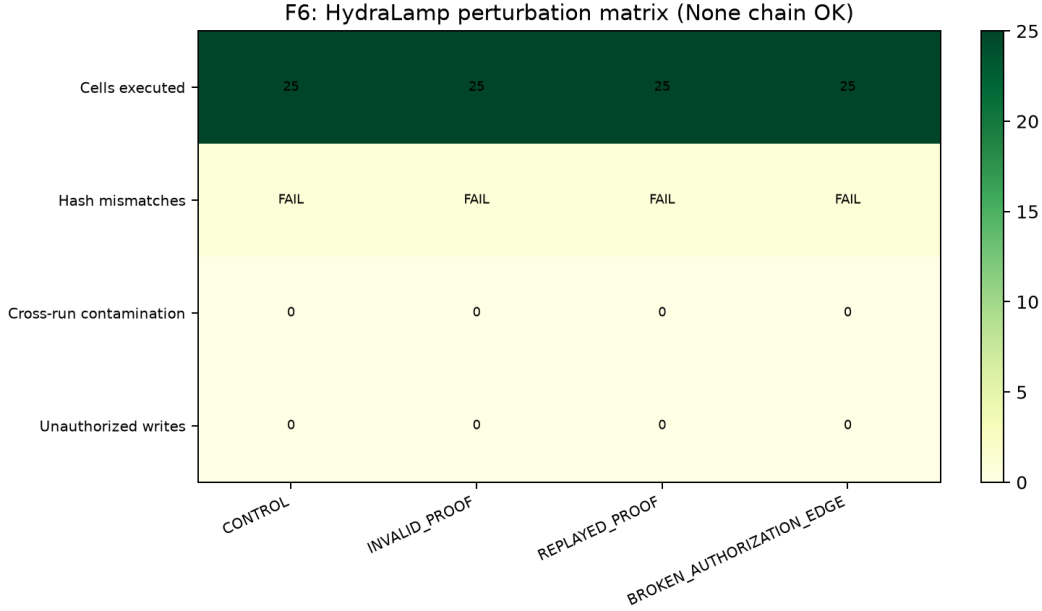


Figure 3: HydraLamp perturbation test matrix (systems robustness only; not treatment-effect evidence).

163 **EXP-009.** Primary verdict remains UNDERPOWERED for confirmatory E06 ordering. An ex-
164 ploratory secondary pattern was recorded as directionally positive at the descriptive layer only; it is
165 **not promoted** to a causal claim.

166 **Stage-2 baseline.** Prior IC Failure Learning stage-2 execution (414 scored rows across two models)
167 established that failure-learning behavior improvement was not established under that frozen design,
168 providing historical context for the structured-retrieval falsification lane. Historical LongMemEval
169 track-03 lanes (local open-weights models, frozen primary scorer) are retained for internal audit
170 only; they were not executed with frontier models and do not upgrade primary EXP-008/009 claims.

171 5.1 Failure-preserving systems validation

172 The preregistered retrieval experiments above test whether structured context improves a frozen end-
173 point. A separate systems-validation suite asks a different question: does the custody mechanism
174 preserve integrity under perturbation, tampering, concurrency, replay, and external execution fail-
175 ure? Table 2 summarizes source-verified outcomes from a frozen HydraLamp perturbation matrix;
176 these results evaluate custody mechanics and do **not** constitute evidence that structured FCG context
177 improves the EXP-008/009 endpoint.

178 The tamper suite used synthetic cases only (not real security incidents). In a live provider repair lad-
179 der, earlier infrastructure gates passed before a structured call hit an external quota limit; the failure
180 remained a terminal auditable state rather than being erased or substituted. The same custody primi-
181 tives have been instantiated in sibling systems for local-model orchestration, deterministic scientific-
182 document atomization, offline anomaly detection, bounded bioinformatics, substrate generation, and
183 operator-facing scientific workflow monitoring; these implementations motivate transfer studies but
184 do not provide additional evidence for the treatment effect reported here.

Table 2: Systems-validation outcomes (custody mechanics only; not treatment-effect evidence).

Validation	Scope	Outcome
Perturbation matrix	100 cells	100/100 chain verification
Synthetic tamper suite	8 modes	8/8 detected
Concurrent execution	10 runs	PASS (10 unique run IDs)
Replay/restart recovery	44 events	PASS
Live provider ladder	R0–R6	Bounded external failure preserved

6 Discussion

The principal contribution of this submission is not evidence that structured FCG retrieval improves failure prevention. EXP-008 and EXP-009 remain *underpowered*; the empirical retrieval effect on the preregistered E06 endpoint is *not established*. The broader methodological proposition is that scientific computational models can expose their evidence-state transitions mechanically—through governed receipts, explicit claim ceilings, and preserved terminal states—so that null, failure, and blocked outcomes remain part of model state rather than disappearing from the public record.

HydraDG demonstrates bounded custody and systems-validation behavior under that contract: failures are first-class, hashes are mandatory, and claim ceilings block over-interpretation. HydraLamp perturbation, tamper, concurrency, and replay results (Table 2) evaluate custody mechanics only. Companion implementation lineages are admitted as related implementations excluded from primary inference; their run receipts are not available in the reviewed source set for this submission. The EXP-008/009 underpowered terminations bound what cannot yet be claimed while preserving malformed and timeout cells for audit.

6.1 Limitations

Limitations include bounded case replication, deliberate local-only small-model lanes in the reported matrix (no frontier API substitution), and absence of runtime-equivalent successor replication in primary results. Execution hardware was constrained to M1-class Apple Silicon with 32 GB RAM; results do not establish performance on datacenter GPU fleets or frontier-model endpoints. A pre-registered Qwen 3.8 successor lane exists but is non-terminal and omitted from primary results. Whole-project hierarchical atomization (SeedGraph v1a validation) ran for multiple days but was interrupted during final Parquet serialization without a terminal build receipt; partial node/edge artifacts are not readback-safe, so terminal whole-corpus atomization is not established in this submission. Biological, protein-structure, and cellular perturbation applications of the custody framework remain future validation targets, not demonstrated generalizations of the agent-context results here.

6.2 Future directions

Agent systems. Larger successor replications, governed multi-agent handoffs, selective replay, and resource-aware custody gates under preregistered manifests.

Structured retrieval at scale. Checkpointed ingest and transactionally durable atomization so long hierarchy builds can resume from batch boundaries rather than materializing the entire graph at finalization.

Cross-project federation. Project-local FCG roots with explicit cross-project references and no automatic evidence admission between repositories.

Cryptographic custody. Authorized signatures and MMR commitments where policy permits; hashes remain identity commitments unless a verified signing operation occurred.

All directions above are *planned* unless independently measured; they do not extend the EXP-008/009 empirical claims.

7 Conclusion

We proposed the term *Mechanical Scientific Model* for a computational model class whose governed state transitions expose evidence, separate deterministic from probabilistic outputs, preserve failure-complete terminal states, and bind explicit claim ceilings with forward and reverse traceability. FCO/FCG provide the custody substrate; HydraDG is one implementation examined here under that proposal. Two preregistered HydraDG studies (EXP-008, EXP-009) closed as *underpowered*, demonstrating bounded custody and systems behavior while withholding confirmatory promotion of structured retrieval effects on failure-prevention endpoints. We recommend explicit Mechanical Scientific Method discipline—custody receipts, claim ceilings, and terminal-state preservation—as baseline infrastructure for NewInML-style agent evaluations, without claiming that structured retrieval efficacy has been established.

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